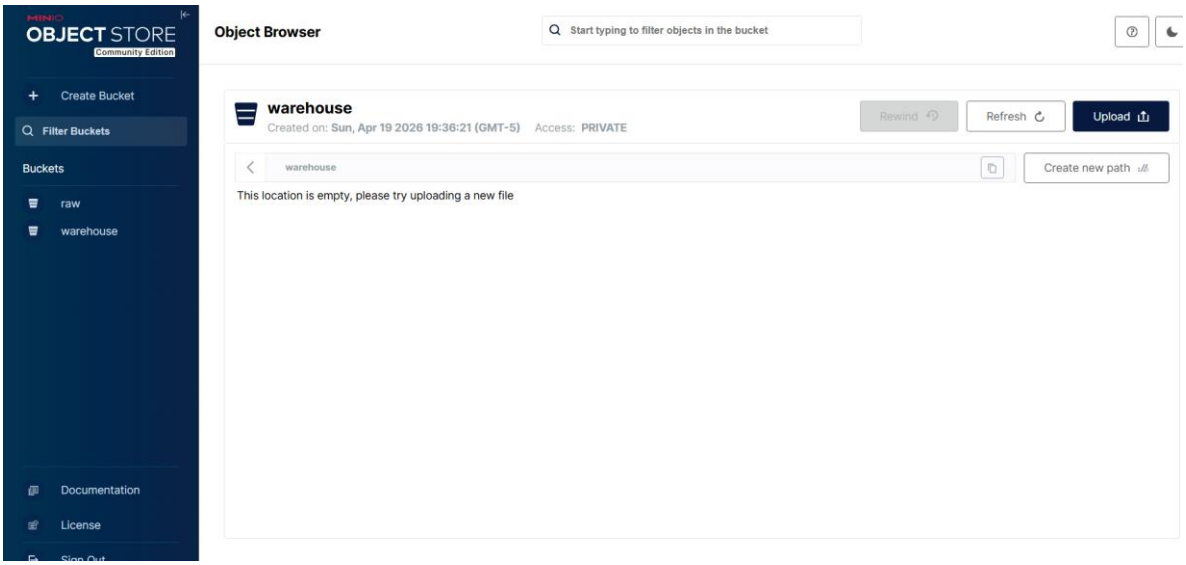
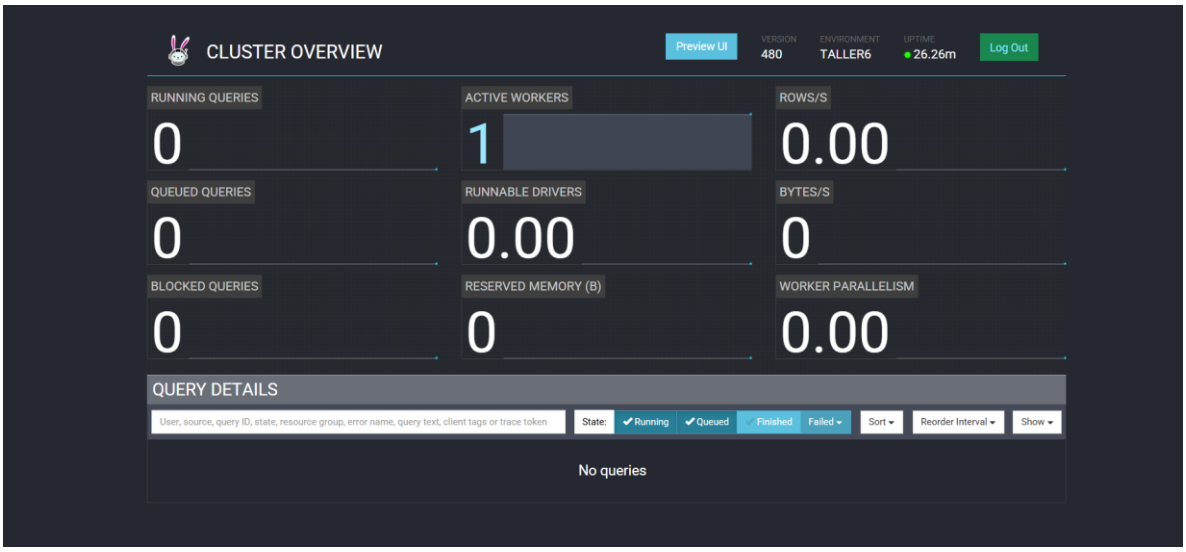





```
PS C:\Users\SAMUEL\Downloads\Talleres-Sistemas-intensivos-master\Taller_6> cat config/trino/catalog/iceberg.properties
connector.name=iceberg
iceberg.catalog.type=rest
iceberg.rest-catalog.uri=http://rest:8181
iceberg.rest-catalog.warehouse=s3://warehouse/
fs.native-s3.enabled=true
s3.endpoint=http://minio:9000
s3.region=us-east-1
s3.path-style-access=true
s3.aws-access-key=admin
s3.aws-secret-key=password123
PS C:\Users\SAMUEL\Downloads\Talleres-Sistemas-intensivos-master\Taller_6>
```

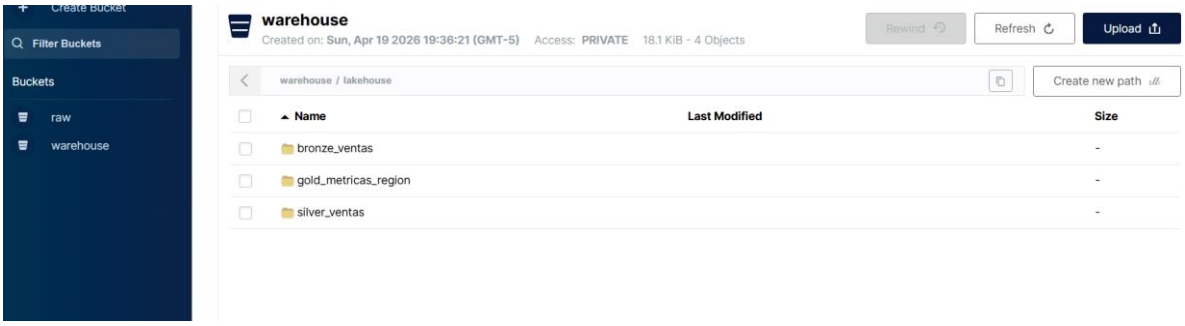
```
[+] Running 63/64L\Downloads\Talleres-Sistemas-intensivos-master\Taller_6> docker compose up -d
 ✔ trino Pulled
 ✔ mc Pulled
 ✔ rest Pulled
 ✔ minio Pulled
 ✔ spark-iceberg Pulled
```

Lab-02

```
Muestra de los primeros 5 registros:
+---+----------+-----------+------------+--------+---------------+------------+--------------------+----------------+
| id|     fecha|   producto|   categoria|cantidad|precio_unitario|      region|        ingestion_ts|     source_file|
+---+----------+-----------+------------+--------+---------------+------------+--------------------+----------------+
|  1|2024-01-05|     Laptop|Computadores|       2|         1200.0|      Bogota|2026-04-20 03:18:...|ventas_lote1.csv|
|  2|2024-01-08|    Teclado| Perifericos|       5|           45.0|    Medellin|2026-04-20 03:18:...|ventas_lote1.csv|
|  3|2024-01-12|    Monitor|   Pantallas|       3|          350.0|      Bogota|2026-04-20 03:18:...|ventas_lote1.csv|
|  4|2024-01-15|      Mouse| Perifericos|       8|           25.0|        Cali|2026-04-20 03:18:...|ventas_lote1.csv|
|  5|2024-01-18|Auriculares|       Audio|       4|           80.0|Barranquilla|2026-04-20 03:18:...|ventas_lote1.csv|
+---+----------+-----------+------------+--------+---------------+------------+--------------------+----------------+
```

```
26/04/20 03:21:59 INFO Executor: Java version 17.0.14
26/04/20 03:21:59 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/04/20 03:21:59 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@7b075028 for default.
26/04/20 03:21:59 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 38361.
26/04/20 03:21:59 INFO NettyBlockTransferService: Server created on 0bfd5733ddc4:38361
26/04/20 03:21:59 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/04/20 03:21:59 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, 0bfd5733ddc4, 38361, None)
26/04/20 03:21:59 INFO BlockManagerMasterEndpoint: Registering block manager 0bfd5733ddc4:38361 with 434.4 MiB RAM, BlockManagerId(driver, 0bfd5
26/04/20 03:21:59 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, 0bfd5733ddc4, 38361, None)
26/04/20 03:21:59 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, 0bfd5733ddc4, 38361, None)
26/04/20 03:22:00 INFO SingleEventLogFileWriter: Logging events to file:/home/iceberg/spark-events/local-1776655319576.inprogress
Leyendo capa Bronze...
  Registros en Bronze: 30
  Registros validos: 30  (eliminados por limpieza: 0)
Cargando dimensiones de productos...

Silver: 30 filas escritas exitosamente.

Muestra Silver (columnas clave):
+---+----------+-----------+------------------+------------+--------+---------------+------+----------+--------+------------+
| id|     fecha|   producto|         proveedor|   categoria|cantidad|precio_unitario| total|margen_pct|ganancia|      region|
+---+----------+-----------+------------------+------------+--------+---------------+------+----------+--------+------------+
|  1|2024-01-05|     Laptop|   TechGlobal S.A.|Computadores|       2|         1200.0|2400.0|      0.32|   768.0|      Bogota|
|  2|2024-01-08|    Teclado|InputDevices Ltda.| Perifericos|       5|           45.0| 225.0|       0.4|    90.0|    Medellin|
|  3|2024-01-12|    Monitor|   VisualPro Corp.|   Pantallas|       3|          350.0|1050.0|      0.28|   294.0|      Bogota|
|  4|2024-01-15|      Mouse|InputDevices Ltda.| Perifericos|       8|           25.0| 200.0|      0.42|    84.0|        Cali|
|  5|2024-01-18|Auriculares|    SoundWave Inc.|       Audio|       4|           80.0| 320.0|      0.35|   112.0|Barranquilla|
+---+----------+-----------+------------------+------------+--------+---------------+------+----------+--------+------------+
```

```
26/04/20 03:24:01 INFO Executor: OS info Linux, 6.6.87.2-microsoft-standard-WSL2, amd64
26/04/20 03:24:01 INFO Executor: Java version 17.0.14
26/04/20 03:24:01 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/04/20 03:24:01 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@7b075028 for defau
26/04/20 03:24:01 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 36483.
26/04/20 03:24:01 INFO NettyBlockTransferService: Server created on 0bfd5733ddc4:36483
26/04/20 03:24:01 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/04/20 03:24:01 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, 0bfd5733ddc4, 36483, None)
26/04/20 03:24:01 INFO BlockManagerMasterEndpoint: Registering block manager 0bfd5733ddc4:36483 with 434.4 MiB RAM, BlockManagerId(d
26/04/20 03:24:01 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, 0bfd5733ddc4, 36483, None)
26/04/20 03:24:01 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, 0bfd5733ddc4, 36483, None)
26/04/20 03:24:01 INFO SingleEventLogFileWriter: Logging events to file:/home/iceberg/spark-events/local-1776655441226.inprogress
Leyendo capa Silver...
  Registros en Silver: 30

Gold: 29 filas de metricas generadas.

Metricas por region:
+------------+----------------+--------------+-----------------+-----------------+
|      region|ingresos_totales|ganancia_total|num_transacciones|unidades_vendidas|
+------------+----------------+--------------+-----------------+-----------------+
|      Bogota|         13668.0|       4102.48|               12|               66|
|    Medellin|          6753.0|       2156.94|                7|               26|
|Barranquilla|          5755.0|        1490.9|                5|               24|
|        Cali|          4101.0|        1173.7|                6|               35|
+------------+----------------+--------------+-----------------+-----------------+
```




```
Total filas Bronze: 30
```

Capa Silver: datos limpios y enriquecidos

Silver agrega valor sobre Bronze de tres formas:

1. **Limpieza**: elimina filas con nulls y valores invalidos.
2. **Enriquecimiento**: join con dimensiones de productos (proveedor, margen).
3. **Calculo**: columnas derivadas `total` y `ganancia`.

[7]:
```python
silver = spark.sql("SELECT * FROM demo.lakehouse.silver_ventas")

print("Schema de Silver:")
silver.printSchema()

print("Primeras 5 filas (columnas clave):")
silver.select(
    "id", "fecha", "producto", "proveedor", "categoria",
    "cantidad", "precio_unitario", "total", "margen_pct", "ganancia", "region"
).show(5)

print(f"Total filas Silver: {silver.count()}")

# Columnas nuevas vs Bronze
bronze_cols = set(spark.sql("SELECT * FROM demo.lakehouse.bronze_ventas LIMIT 1").columns)
silver_cols = set(silver.columns)
nuevas = silver_cols - bronze_cols
print(f"\nColumnas nuevas en Silver (no existen en Bronze): {sorted(nuevas)}")
```

```
Schema de Silver:
root
 |-- producto: string (nullable = true)
 |-- id: integer (nullable = true)
 |-- fecha: date (nullable = true)
 |-- categoria: string (nullable = true)
 |-- cantidad: integer (nullable = true)
 |-- precio_unitario: double (nullable = true)
 |-- region: string (nullable = true)
 |-- ingestion_ts: timestamp (nullable = true)
 |-- source_file: string (nullable = true)
 |-- proveedor: string (nullable = true)
 |-- margen_pct: double (nullable = true)
 |-- total: double (nullable = true)
 |-- ganancia: double (nullable = true)
 |-- silver_ts: timestamp (nullable = true)

Primeras 5 filas (columnas clave):
+---+----------+----------+------------------+-----------+--------+----------------+------+----------+--------+-----------+
| id|     fecha|  producto|         proveedor|  categoria|cantidad|precio_unitario| total|margen_pct|ganancia|     region|
+---+----------+----------+------------------+-----------+--------+----------------+------+----------+--------+-----------+
|  1|2024-01-05|    Laptop|   TechGlobal S.A.|Computadores|       2|          1200.0|2400.0|      0.32|   768.0|     Bogota|
|  2|2024-01-08|   Teclado|InputDevices Ltda.|  Perifericos|       5|            45.0| 225.0|       0.4|    90.0|   Medellin|
|  3|2024-01-12|   Monitor|   VisualPro Corp.|   Pantallas|       3|           350.0|1050.0|      0.28|   294.0|     Bogota|
|  4|2024-01-15|     Mouse|InputDevices Ltda.|  Perifericos|       8|            25.0| 200.0|      0.42|    84.0|       Cali|
|  5|2024-01-18|Auriculares|    SoundWave Inc.|      Audio|       4|            80.0| 320.0|      0.35|   112.0|Barranquilla|
+---+----------+----------+------------------+-----------+--------+----------------+------+----------+--------+-----------+
only showing top 5 rows
```

Lab-03

T6 MinIO (S3) Spark + Trino Iceberg Lakehouse: capas, catalogo compa

Reflexion final: ¿Que habria que cambiar si el storage fuera GCS o Azure Blob en lugar de MinIO?

[5]:
```
bronze = spark.sql("SELECT * FROM demo.lakehouse.bronze_ventas")
print(f"Total filas Bronze: {bronze.count()}")

spark.sql("""
    SELECT snapshot_id, committed_at, operation
    FROM demo.lakehouse.bronze_ventas.snapshots
""").show()
```

```
Total filas Bronze: 30
+-------------------+--------------------+---------+
|        snapshot_id|        committed_at|operation|
+-------------------+--------------------+---------+
|1670595596177227778|2026-04-20 03:53:...|   append|
+-------------------+--------------------+---------+
```

```
from pyspark.sql import functions as F

# Leer lote 2 (20 filas Q2 2024)
df_lote2 = spark.read.csv("/home/iceberg/data/ventas_lote2.csv", header=True, inferSchema=True)

# Agregar columnas de auditoría
df_lote2 = (
    df_lote2
    .withColumn("ingestion_ts", F.current_timestamp())
    .withColumn("source_file", F.lit("ventas_lote2.csv"))
)

# APPEND a Bronze (no sobrescribe)
df_lote2.writeTo("demo.lakehouse.bronze_ventas").append()

# Verificar total
total_bronze = spark.sql("SELECT COUNT(*) FROM demo.lakehouse.bronze_ventas").collect()[0][0]
print(f"Total filas en Bronze ahora: {total_bronze}")  # debe ser 50
```

```
Total filas en Bronze ahora: 50
```

7]:
```
spark.sql("""
    SELECT snapshot_id,
           committed_at,
           operation,
           summary['added-records'] AS registros_agregados,
           summary['total-records'] AS total_registros
    FROM demo.lakehouse.bronze_ventas.snapshots
    ORDER BY committed_at
""").show(truncate=False)
```

```
+-------------------+----------------------+---------+-------------------+---------------+
|snapshot_id        |committed_at          |operation|registros_agregados|total_registros|
+-------------------+----------------------+---------+-------------------+---------------+
|1670595596177227778|2026-04-20 03:53:28.956|append   |30                 |30             |
|5186926011069854 22|2026-04-20 03:56:01.223|append   |20                 |50             |
+-------------------+----------------------+---------+-------------------+---------------+
```

```
26/04/20 03:57:07 INFO Executor: Java version 17.0.14
26/04/20 03:57:07 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/04/20 03:57:07 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@1b8e3389 for default.
26/04/20 03:57:07 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 40525.
26/04/20 03:57:07 INFO NettyBlockTransferService: Server created on ac450d5d260f:40525
26/04/20 03:57:07 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/04/20 03:57:07 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, ac450d5d260f, 40525, None)
26/04/20 03:57:07 INFO BlockManagerMasterEndpoint: Registering block manager ac450d5d260f:40525 with 434.4 MiB RAM, BlockManagerId(driver, ac450d5d260f, 40525,
26/04/20 03:57:07 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, ac450d5d260f, 40525, None)
26/04/20 03:57:07 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, ac450d5d260f, 40525, None)
26/04/20 03:57:07 INFO SingleEventLogFileWriter: Logging events to file:/home/iceberg/spark-events/local-1776657426975.inprogress
Leyendo capa Bronze...
  Registros en Bronze: 50
  Registros validos: 50  (eliminados por limpieza: 0)
Cargando dimensiones de productos...

Silver: 50 filas escritas exitosamente.

Muestra Silver (columnas clave):
+---+----------+-----------+------------------+------------+--------+---------------+------+----------+--------+------------+
| id|     fecha|   producto|         proveedor|   categoria|cantidad|precio_unitario| total|margen_pct|ganancia|      region|
+---+----------+-----------+------------------+------------+--------+---------------+------+----------+--------+------------+
|  1|2024-01-05|     Laptop|   TechGlobal S.A.|Computadores|       2|         1200.0|2400.0|      0.32|   768.0|      Bogota|
|  2|2024-01-08|    Teclado|InputDevices Ltda.| Perifericos|       5|           45.0| 225.0|       0.4|    90.0|    Medellin|
|  3|2024-01-12|    Monitor|   VisualPro Corp.|   Pantallas|       3|          350.0|1050.0|      0.28|   294.0|      Bogota|
|  4|2024-01-15|      Mouse|InputDevices Ltda.| Perifericos|       8|           25.0| 200.0|      0.42|    84.0|        Cali|
|  5|2024-01-18|Auriculares|    SoundWave Inc.|       Audio|       4|           80.0| 320.0|      0.35|   112.0|Barranquilla|
```

```
26/04/20 03:57:47 INFO Executor: OS info Linux, 6.6.87.2-microsoft-standard-WSL2, amd64
26/04/20 03:57:47 INFO Executor: Java version 17.0.14
26/04/20 03:57:47 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/04/20 03:57:47 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@6e09bb75 for default.
26/04/20 03:57:47 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 34815.
26/04/20 03:57:47 INFO NettyBlockTransferService: Server created on ac450d5d260f:34815
26/04/20 03:57:47 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/04/20 03:57:47 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, ac450d5d260f, 34815, None)
26/04/20 03:57:47 INFO BlockManagerMasterEndpoint: Registering block manager ac450d5d260f:34815 with 434.4 MiB RAM, BlockManagerId(driver, ac
26/04/20 03:57:47 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, ac450d5d260f, 34815, None)
26/04/20 03:57:47 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, ac450d5d260f, 34815, None)
26/04/20 03:57:48 INFO SingleEventLogFileWriter: Logging events to file:/home/iceberg/spark-events/local-1776657467259.inprogress
Leyendo capa Silver...
  Registros en Silver: 50

Gold: 48 filas de metricas generadas.

Metricas por region:
+------------+----------------+------------------+----------------+-----------------+
|      region|ingresos_totales|    ganancia_total|num_transacciones|unidades_vendidas|
+------------+----------------+------------------+----------------+-----------------+
|      Bogota|         28684.0|           8557.48|              21|              120|
|        Cali|          9997.0|            2684.7|              10|               54|
|    Medellin|          8919.0|           2871.96|              11|               55|
|Barranquilla|          8895.0|2417.7000000000003|               8|               42|
+------------+----------------+------------------+----------------+-----------------+
```

Lab – 04

```
PS C:\Users\SAMUEL\Downloads\Talleres-Sistemas-intensivos-master\Taller_6> docker compose exec trino trino --catalog iceberg
f region ORDER BY ingresos DESC"
time="2026-04-19T23:01:02-05:00" level=warning msg="C:\\Users\\SAMUEL\\Downloads\\Talleres-Sistemas-intensivos-master\\Talle
tial confusion"
"Bogota","28684.0"
"Cali","9997.0"
"Medellin","8919.0"
"Barranquilla","8895.0"
PS C:\Users\SAMUEL\Downloads\Talleres-Sistemas-intensivos-master\Taller_6>
```

```
PS C:\Users\SAMUEL\Downloads\Talleres-Sistemas-intensivos-master\Ta
time="2026-04-19T23:02:00-05:00" level=warning msg="C:\\Users\\SAMU
tial confusion"
trino:lakehouse> - SHOW TABLES;
        Table
----------------------
 bronze_ventas
 gold_metricas_region
 silver_ventas
(3 rows)

Query 20260420_040217_00003_24tj2, FINISHED, 1 node
Splits: 7 total, 7 done (100.00%)
0.23 [3 rows, 305B] [12 rows/s, 1.27KiB/s]

trino:lakehouse>
```

```
NT', 'INSERT', 'MERGE', 'PREPARE', 'REFRESH', 'RESET', 'REVOKE', 'ROLLBACK', 'SET', 'SHOW', 'START', 'TRUNCATE', 'UPDATE', 'USE', 'WITH', <query>
s_region

trino:lakehouse> DESCRIBE gold_metricas_regioas_region;
Query 20260420_040451_00007_24tj2 failed: line 1:1: Table 'iceberg.lakehouse.gold_metricas_regioas_region' does not exist
DESCRIBE gold_metricas_regioas_region

trino:lakehouse> DESCRIBE gold_metricas_region;
      Column       |            Type            | Extra | Comment
-------------------+----------------------------+-------+---------
 region            | varchar                    |       |
 categoria         | varchar                    |       |
 anio              | integer                    |       |
 mes               | integer                    |       |
 anio_mes          | varchar                    |       |
 ingresos_totales  | double                     |       |
 ganancia_total    | double                     |       |
 num_transacciones | bigint                     |       |
 ticket_promedio   | double                     |       |
 unidades_vendidas | bigint                     |       |
 gold_ts           | timestamp(6) with time zone |       |
(11 rows)

Query 20260420_040509_00008_24tj2, FINISHED, 1 node
Splits: 7 total, 7 done (100.00%)
0.23 [11 rows, 2.13KiB] [48 rows/s, 9.4KiB/s]
```

```
0.23 [11 rows, 2.13KiB] [48 rows/s, 9.4KiB/s]

trino:lakehous   SELECT region,
              ->        SUM(ingresos_totales) AS ingresos_totales,
              ->        SUM(ganancia_total) AS ganancia_total,
              ->        SUM(num_transacciones) AS num_transacciones
              -> FROM gold_metricas_region
              -> GROUP BY region
              -> ORDER BY ingresos_totales DESC;SELECT region,
    region    | ingresos_totales |   ganancia_total   | num_transacciones
--------------+------------------+--------------------+-------------------
 Bogota       |          28684.0 |            8557.48 |                21
 Cali         |           9997.0 |             2684.7 |                10
 Medellin     |           8919.0 |            2871.96 |                11
 Barranquilla |           8895.0 | 2417.7000000000003 |                 8
(4 rows)

Query 20260420_040538_00009_24tj2, FINISHED, 1 node
Splits: 15 total, 15 done (100.00%)
0.43 [48 rows, 5.22KiB] [112 rows/s, 12.3KiB/s]
```

```
;                SELECT categoria,
              ->        SUM(ingresos_totales) AS ingresos,
              ->        COUNT(*) AS num_periodos
              -> FROM gold_metricas_region
              -> GROUP BY categoria
              -> ORDER BY ingresos DESC;
   categoria     | ingresos | num_periodos
----------------+----------+--------------
 Moviles         |  19480.0 |            8
 Computadores    |  18110.0 |            7
 Pantallas       |   5140.0 |            5
 Perifericos     |   4821.0 |           13
 Almacenamiento  |   3640.0 |            5
 Componentes     |   3162.0 |            5
 Audio           |   2142.0 |            5
(7 rows)

Query 20260420_040742_00013_24tj2, FINISHED, 1 node
Splits: 15 total, 15 done (100.00%)
0.29 [48 rows, 5.5KiB] [163 rows/s, 18.8KiB/s]
```

```
Splits: 15 total, 15 done (100.00%)
0.29 [48 rows, 5.5KiB] [163 rows/s, 18.8KiB/s]

trino:lakehouse> SELECT anio_mes,
               ->        SUM(ingresos_totales) AS ingresos_mes
               -> FROM gold_metricas_region
               -> GROUP BY anio_mes
               -> ORDER BY anio_mes;
 anio_mes | ingresos_mes
----------+--------------
 2024-01  |       5715.0
 2024-02  |       8323.0
 2024-03  |      16239.0
 2024-04  |      10331.0
 2024-05  |      10180.0
 2024-06  |       5707.0
(6 rows)

Query 20260420_040800_00014_24tj2, FINISHED, 1 node
Splits: 15 total, 15 done (100.00%)
0.25 [48 rows, 5.05KiB] [190 rows/s, 20.1KiB/s]
```

```
Query 20260420_040800_00014_24tj2, FINISHED, 1 node
Splits: 15 total, 15 done (100.00%)
0.25 [48 rows, 5.05KiB] [190 rows/s, 20.1KiB/s]

trino:lakehouse> SELECT producto,
               ->        COUNT(*) AS num_ventas,
               ->        SUM(total) AS ingresos,
               ->        SUM(ganancia) AS ganancia_total
               -> FROM silver_ventas
               -> GROUP BY producto
               -> ORDER BY ingresos DESC
               -> LIMIT 5;
  producto  | num_ventas | ingresos | ganancia_total
------------+------------+----------+----------------
 Laptop     |          7 |  18110.0 |         5795.2
 Smartphone |          5 |  15480.0 |         3405.6
 Monitor    |          5 |   5140.0 |         1439.2
 Tablet     |          3 |   4000.0 |         1080.0
 Disco SSD  |          5 |   3640.0 |         1092.0
(5 rows)

Query 20260420_040811_00015_24tj2, FINISHED, 1 node
Splits: 19 total, 19 done (100.00%)
0.38 [50 rows, 6.57KiB] [130 rows/s, 17.2KiB/s]
```

```
 Disco SSD  |          5 |   3640.0 |         1092.0
(5 rows)

Query 20260420_040811_00015_24tj2, FINISHED, 1 node
Splits: 19 total, 19 done (100.00%)
0.38 [50 rows, 6.57KiB] [130 rows/s, 17.2KiB/s]

trino:lakehouse> SELECT CASE
               ->          WHEN CAST(fecha AS DATE) BETWEEN DATE '2024-01-01' AND DATE '2024-03-31' THEN 'Q1'
               ->          WHEN CAST(fecha AS DATE) BETWEEN DATE '2024-04-01' AND DATE '2024-06-30' THEN 'Q2'
               ->        END AS trimestre,
               ->        COUNT(*) AS num_ventas,
               ->        SUM(total) AS ingresos_totales,
               ->        AVG(ganancia) AS ganancia_promedio_venta
               -> FROM silver_ventas
               -> GROUP BY 1
               -> ORDER BY 1;
 trimestre | num_ventas | ingresos_totales | ganancia_promedio_venta
-----------+------------+------------------+------------------------
 Q1        |         30 |          30277.0 |      297.46733333333333
 Q2        |         20 |          26218.0 |      380.39099999999996
(2 rows)

Query 20260420_040829_00016_24tj2, FINISHED, 1 node
Splits: 15 total, 15 done (100.00%)
```